

Lab 1 Solutions

Statistics for Laboratory Scientists II

1. Preparations:

```
# data
x <- read.csv("data1-1.csv")
# just need second column
x <- x[,2]
```

```
# expected ratios
p <- c(18,6,6,2,12,4,12,4)
# rescale to sum to 1
p <- p/sum(p)
```

```
# expected counts
ex <- sum(x)*p
```

(a) The statistics:

```
# chi-square test statistic
chi <- sum((x-ex)^2/ex) # value = 9.53

# LRT statistic
lrt <- 2*sum(x*log(x/ex)) # value = 8.82
```

(b) The asymptotic P-values (df = no. cells – 1 = 7):

```
1 - pchisq(chi, 7) # p-value = 0.22
1 - pchisq(lrt, 7) # p-value = 0.27
```

(c) Simulation-based P-values:

```
n.sim <- 1000
n <- sum(x)
results <- matrix(ncol=2,nrow=n.sim)
for(i in 1:n.sim) {
  simdat <- rmultinom(1,n,p)
  results[i,1] <- sum((simdat-ex)^2/ex)
  results[i,2] <- 2*sum(simdat[simdat!=0]*log(simdat[simdat!=0]/ex))
}
mean(results[,1] >= chi) # p-value = 0.21
mean(results[,2] >= lrt) # p-value = 0.27
```

(d) Conclusions: As the P-values are large, we fail to reject the null hypothesis. The observed data conform reasonably to the hypothesized proportions.

2. Preparations:

```
y <- read.csv("data1-2.csv")
# just need the last two columns
y <- y[,4:5]
```

(a) The chi-square and likelihood ratio tests:

```
# chi-square statistic: use chisq.test()
chi <- chisq.test(y) # statistic = 7.25; P-value = 0.40
```

```
# LRT
ex <- chi$expected # expected counts
lrt <- 2*sum(y * log(y/ex)) # statistic = 7.24
1 - pchisq(lrt, nrow(y)-1) # p-value = 0.40
```

(b) Fisher's exact test:

```
fisher.test(y) # p-value = 0.43
```

(c) Conclusions: As the P-values are large, we fail to reject the null hypothesis. There is no evidence for a difference, among the conditions, in the frequency with which the ticks choose the treated tube.